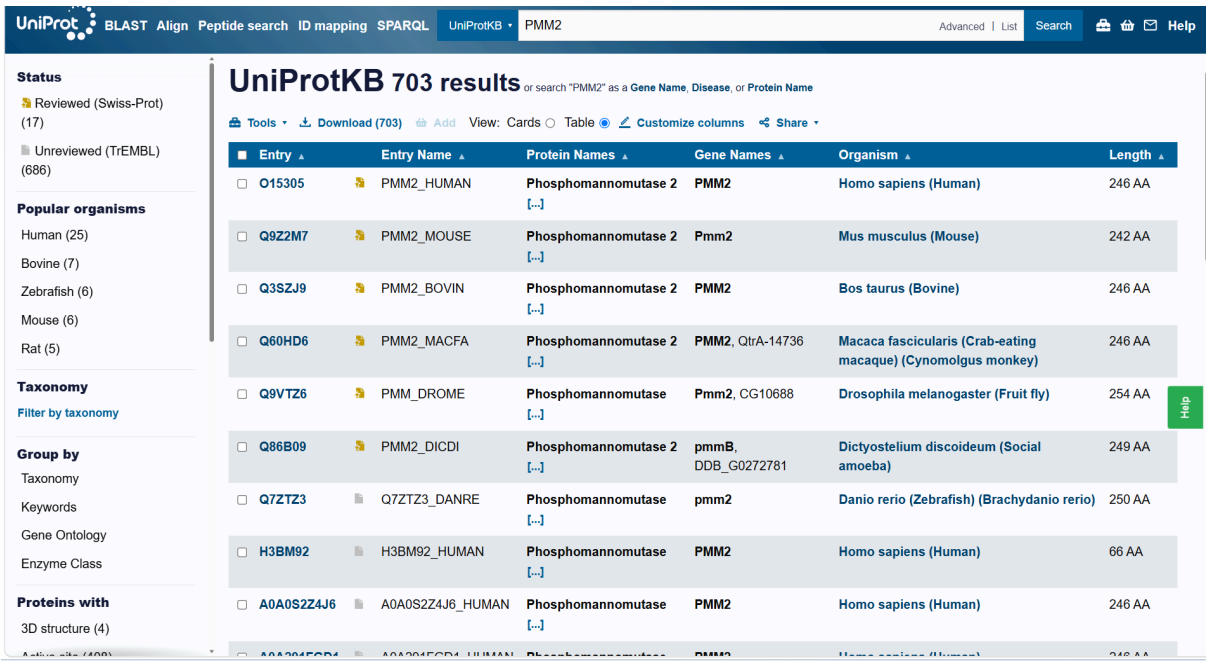


PMM2 -HOMO SAPIENS

>sp|O15305|PMM2_HUMAN Phosphomannomutase 2 OS=Homo sapiens OX=9606 GN=PMM2 PE=1 SV=1
MAAPGPALCLFDVDGTLTAPRQKITKEMDDFLQKLRQKIKIGVVGGSDFEKVQEQLGNDV
VEKYDYVFPENGLVAYKDGKLLCRQNIQSHLGEALIQDLINYCLSYIAKIKLPKKRGTFI
EFRNGMLNVSPIGRSCSQEERIEFYELDKKENIRQKFVADLRKEFAGKGLTFSIGGQISF
DVFPDGWDKRYCLRHVENDGYKTIYFFGDKTMPGGNDHEIFTDPRTMGYSVTAPEDTRRI
CELLFS

PPM2- LEISHMANIA MEXICANA

>sp|Q4Q1M7|PMM_LEIMA Phosphomannomutase OS=Leishmania major OX=5664 GN=pmm PE=3 SV=1
MTSKVILLFDVDGTLTPPRNPETQDMKETLAKARAAGFKLGVVGGSDFAKQKEQLGDSIL
EDFDYVFSENGLLAYKDGTEFHRNSLLKALGNEKVVAFVKKCLHLIADLDIPVQRGTFVE
FRNGMFNVSPIGRNCSQQERDEFEQLDKERHIREKLIRNLKEAFPDYPLAYSVGGQISFD
VFPKGWDKTYCLQFVENDFEKIHFFGDKTSEGGNDYEIYTDCRTIGHSVKTYKDTIAILE
ALLEESR

PPM2- VICUGNA PACOS

>tr|A0A6I9I8N5|A0A6I9I8N5_VICPA Phosphomannomutase OS=Vicugna pacos OX=30538 GN=PMM2 PE=3 SV=1

MAAPRPALCLFDVDGTLTAPRQKITKDMDGFLQELRQKIKIGVVGGSDFEKVQEQLGSDV
VEKYDYVFPENGLVAYKDGKLLCKQNIQGHLGEALIQDLINYCLSYIAKIKLPRKRGTFI
EFRNGMLNVSPVGRSCSQEERIEFYELDQKENIRQKFVEDLRKEFAGKGLTFSIGGQISF
DVFPDGWDKRYCLRHVENDGYKTIYFFGDKTMPGGNDHEIFTDPRTVGYTVTAPDDTRRI
CQELFC

Aligning the sequences with different species allows us to examine the sequences of the gene and determine if the mutation is important by locating it in other species. As evolution and natural selection made that mutation continued.

If the mutation in homo sapiens is located in other species it means that it could be important, it has a beneficial function and importance in the structure of the protein or its activity. Having that amino acid was relevant for the structure that a couple of species have the same.

If the other species have the wild-type amino acid a change might bring out a malfunction,affect the stability of the protein and cause the disease. However, we would need further investigation to determine certainly if it brings the disease or not.